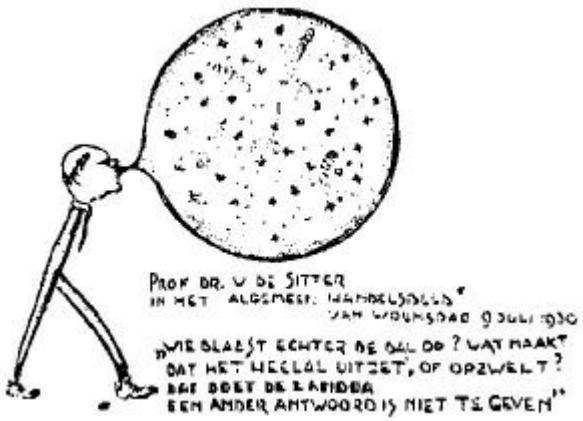
7. See P.J.E. Peebles, Principles of Physical Cosmology (Princeton: Princeton University Press, 1993), p. 81.



The caption reads: “But who is really blowing up this ball? What makes it so that the universe expands or inflates? A Lambda does the job! Another answer cannot be given.” (Translation by Koenraad Schalm.) Lambda refers to something known as the cosmological constant, an idea we will encounter in Chapter 10.

8. To avoid confusion, let me note that one drawback of the penny model is that every penny is essentially identical to every other, while that is certainly not true of galaxies. But the point is that on the largest of scales—scales on the order of 100 million light-years—the individual differences between galaxies are believed to average out so that, when one analyzes huge volumes of space, the overall properties of each such volume are extremely similar to the properties of any other such volume.

9. You could also travel to just outside the edge of a black hole, and remain there, engines firing away to avoid being pulled in. The black hole’s strong gravitational field manifests itself as a severe warping of spacetime, and that results in your clock’s ticking far slower than it would in a more ordinary location in the galaxy (as in a relatively empty spatial expanse). Again, the time duration measured by your clock is perfectly valid. But, as in zipping around at high speed, it is a completely individualistic perspective. When analyzing features of the universe as a whole, it is more useful to have a widely applicable and agreed upon notion of elapsed time, and that’s what is provided by clocks that move along with the cosmic flow of spatial expansion and that are subject to a far more mild, far more average gravitational field.